Project Proposal Report

Date	06 July 2026	**Project Name**	Smart Lender
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Project Proposal (Proposed Solution)

Project Overview

The primary objective is to revolutionize the loan approval process by implementing advanced machine learning techniques, ensuring faster and more accurate assessments. The project comprehensively assesses and enhances the loan approval process, incorporating machine learning for a more robust and efficient system.

Problem Statement Impact

Addressing inaccuracies and inefficiencies in the current loan approval system adversely affects operational efficiency and customer satisfaction. Solving these issues will result in improved operational efficiency, reduced risks, and an overall enhancement in the lending process, contributing to customer satisfaction.

Resource Requirements Specification

Resource Type	Description	Specification/Allocation
Hardware	Computing Resources	T4 GPU, 8 Cores CPU
Hardware	Memory (RAM)	8 GB RAM minimum configuration
Hardware	Storage space	1 TB SSD storage
Software	Web Framework	Python 3.13, Flask Framework
Software	Libraries	scikit-learn, xgboost, imbalanced-learn, pandas, numpy
Software	IDE	Jupyter Notebook, VS Code, PyCharm
Data	Data Source	Kaggle loan prediction dataset (614 rows, CSV format)